REPUBLIC POLYTECHNIC

C339 Software Testing and Analytics
Continuous Assessment 1 (CA1)

Project 02: Library Loan System Requirements Specification

1. About

A library requires a simple software system to calculate overdue fines and determine whether a reader may borrow another book.

The system consists of two independent functions:

- **Calculate Overdue Fine:** Calculate an overdue fine from the number of overdue days.
- **Determine Borrowing Eligibility:** Determine whether a reader is allowed to borrow one more book.

Both functions must operate according to the requirements described below.

2 General Requirements

The following requirements apply to both functions:

1. All monetary amounts, where used, are represented in Singapore cents (SGD), using whole numbers without decimal places.
2. Numeric inputs must be Python integers (int). Boolean values are not valid integers. A parameter explicitly declared as bool accepts only True or False.
3. If an argument has an invalid data type, the function must raise a **TypeError**.
4. If an argument has the correct data type but falls outside the permitted range, the function must raise a **ValueError**.
5. The functions operate independently and do not require a database, network connection, or shared application state.

3 Functional Requirements

3.1 Calculate Overdue Fine

Source File: library_fine.py

Function: calculate_fine(days_overdue)

Description

Calculate an overdue fine from the number of overdue days.

Parameter	Data Type	Constraints
days_overdue	int	Must be a nonnegative integer

Requirements

1. The fine is 50 cents multiplied by the number of overdue days.
2. Zero overdue days results in a fine of zero. There is no maximum fine.

Output: Return the calculated fine as an integer in cents.

3.2 Determine Permit Validity

Source File: library_eligibility.py

Function: can_borrow(active_loans, overdue_books)

Description

Determine whether a reader is allowed to borrow one more book.

Parameter	Data Type	Constraints
active_loans	int	Must be a nonnegative integer
overdue_books	int	Must be a nonnegative integer

Requirements

1. A reader may borrow only when active_loans is less than 5 and overdue_books is zero.
2. A reader with 5 or more active loans, or any overdue books, may not borrow.

Output: Return True when borrowing is allowed and False otherwise. Return a Python bool.